# Staircase Traversal

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# The problem

- Ways to climb `height` steps taking 1..maxSteps at a time.
- Sliding-window sum DP.

# Intuition

- Ways to reach step n is the sum of ways for the last maxSteps steps, a sliding window.

# Approach

- 1. ways of zero is one
- 2. Keep a running window sum
- 3. Add the new value and drop the oldest

# Java solution

```java
public class StaircaseTraversal {
    public static int staircaseTraversal(int height, int maxSteps) {
        int[] ways = new int[height + 1];
        ways[0] = 1;
        int window = 0;
        for (int h = 1; h <= height; h++) {
            window += ways[h - 1];
            if (h - maxSteps - 1 >= 0) window -= ways[h - maxSteps - 1];
            ways[h] = window;
        }
        return ways[height];
    }
}
```

# Complexity

- Time: O(n)
- Space: O(n)

# Recap

- Ways to reach step n is the sum of ways for the last maxSteps steps, a sliding window.
- Complexity: O(n)